Degrees and Qualifications:

BSc (hons) Physiotherapy first class honours with award for clinical placements

Present and previous positions:

2008: Band 5 rotations: Milton Keynes Hospital

2009: Band 5 rotations: Guys and St Thomas Hospital

2010: Band 6 rotations: Guys and St Thomas Hospital

2013: Band 7 Paediatric Respiratory: Evelina London

2014: Band 7 Paediatric Respiratory: Royal London Hospital

2016: Band 7 Paediatric Respiratory: Great Ormond Street Hospital

2023: Band 7 Paediatric Research Respiratory: Great Ormond Street Hospital

2023: Band 8b Workforce Planning: Great Ormond Street Hospital

Publications:

Edel L, Grime C, Robinson V, Manzur A, Abel F, Munot P, Ridout D, Scoto M, Muntoni F, Chan E. A new respiratory scoring system for evaluation of respiratory outcomes in children with spinal muscular atrophy type1 (SMA1) on SMN enhancing drugs. Neuromuscul Disord. 2021 Apr;31(4):300-309.

R. Tillmann, M. Main, L. Abbot, L. Edel, M. Scoto, F. Muntoni (2018) Spinal Muscular Atrophy (SMA) type 1, a changing phenotype: Implications for motor function and physiotherapy management from the Nusinersen Expanded Access Program (EAP). Journal of the APCP.

A. LoMauro, C. Mastella , M. Scoto, F. Muntoni, E. Chan, L. Edel, A. Aliverti and G. Baranello. Can breathing pattern assessment predict the need of ventilatory support in treated SMA1 Infants? Am J Respir Crit Care Med. 2022. Apr.

M. Scoto, M. Main, P. Munot, L. Edel, F. Abel, E. Chan, A. Manzur and F. Muntoni. (2018) Nusinersen is the first drug approved for SMA: Initial Experience in patients with SMA Type1 treated in the EAP. ABSTRACT. Arch Dis Child. 2018. March; 103

C. Lillien, G. Baranello, G. Coratti, L. Edel, O. Germanenko, M. Krstic, M. Mazurkiewicz-Bełdzińska, S. Ray, A. Shatillo, J. Taytard, D. Vlodavets, C. Vuillerot, L. Cruz, G. Tachibana, C. Viljoen and L. Servais. Improving the implementation of spinal muscular atrophy (SMA) standards of care (SoC) internationally – the SMA SoC for all platform. ABSTRACT. Neuromuscul Disord. 2021. Oct; 31(1): S129

G. O'Connor, L. Edel, S. Raquq, M. Bowerman, A. Szmurlo, Z. Simpson, I. Hardy, M. Fewtrell and G. Baranello. Open-labelled study to monitor the effect of an amino acid formula on symptom management in children with spinal muscular atrophy type 1: The SMAAF pilot study. 2023. Aug; 31 (4): 300-309

Childs A, Turner C, Astin R, Bianchi S, Bourke J, Cunningham V, Edel L, Edwards C, Farrant P, J Heraghty, M James, C Massey, B Messer, JM Sodhi, PB Murphy, M Schiava, A Thomas, F Trucco and M Guglieri. Development of respiratory care guidelines for Duchenne muscular dystrophy in the UK: key recommendations for clinical practice. Thorax. ARTICLE IN PRESS

Relevant Prizes, Awards and Academic Distinctions

November 2023	NIHR BRC GOSH Pre Doctoral Internship
	PPIE funding award
October 2018	Lea Rose Spinal Muscular Atrophy Prize for the Best Presentation on SMA

Applicant Research Background (approx. 1000 words – Current 1309 words)

Professional Background: Allied Health Professional

Profession: Physiotherapist

Clinical Development

Following my qualification as a physiotherapist in 2008 and gaining experience across the full range of rotations, adults and paediatrics, I quickly found my passion for paediatric respiratory physiotherapy. I have a passion for the health and wellbeing of children and having the long-term relationships with children and families to enable them to stay well in the community and at home. Education in respiratory management for families and staff to further improve quality of life is a key strategy.

Working in specialist children's hospital services over the last 12 years has expanded my experience, knowledge and clinical expertise in caring for children. During this time, I developed a keen interest in respiratory physiotherapy in neuromuscular conditions. Respiratory compromise in these children has been widely under-recognised and under researched, however it accounts for the majority of hospital admissions, and it is the primary cause of mortality. I aim to develop effective interventions and care plans with children and families to enable longer healthier lives and maximise quality of life.

In 2016 I was appointed to develop and lead the new respiratory neuromuscular service. The neuromuscular service at Great Ormond Street Hospital is the largest in the UK seeing patients through new diagnoses, ongoing management and end-of-life care. Care is provided in a variety of settings, including intensive care, inpatient wards and outpatients for children who are referred from all over the United Kingdom.

My clinical practice is focused on respiratory care from diagnosis through to transition, ensuring patients and their families are supported throughout their journey, over the course of their disease. A key innovation in this role has been developing and leading the service for mechanical insufflation exsufflation (MIE). MIE is a device used by children with muscle weakness to help them produce an effective cough, in order for them to clear mucus and prevent chest infections.

I work alongside the MD UK and SMA UK and regularly help to provide education provision for respiratory neuromuscular physiotherapy, including participation in special interest groups, national and international meetings. Recognising the need for clinicians to share expertise in this rapidly developing field, I have built and led a network of physiotherapists for respiratory care in neuromuscular conditions providing teaching and collaboration. I also lecture at University College London MSc Physiotherapy on respiratory physiotherapy and airway clearance techniques. I have been invited to speak internationally on respiratory care for spinal muscular atrophy (SMA) to a wide range of the multi-disciplinary team, further increasing my opportunities for collaboration.

My special interest in respiratory care for children with SMA developed in response to both clinical need and a recognition that the main outcome assessments for drug trials only focused on whether children were ventilated and for how long. This approach fails to capture other aspects of respiratory management, which may have both clinical and care relevance to children, including hospital attendances and burden of care. This fuels my determination to focus all aspects of this care and ensure we can monitor and support children optimising outcomes.

My experience as a clinician and my desire to improve the care of children with neuromuscular disease, leads me the first stage of a clinical academic career and the PhD would be the first steppingstone in what I plan as a long journey in research. It is my ambition to progress through clinical academia to continue to help change the lives of children with life limiting conditions.

Research Experience

I have had an interest in research throughout my career. Since coming to Great Ormond Street Hospital this interest has developed into a passion, which I have been able to more fully realise as part of a research active physiotherapy department and with the neuromuscular service linked to the UCL Institute of Child Health forming the Dubowitz Neuromuscular Centre of Excellence. I was able to become part of large

research trials by becoming a sub investigator in interventional and national history trials (INCEPTUS, STR1VE-EU, SPR1NT).

Whilst at Great Ormond Street Hospital, I have become increasingly interested in respiratory outcome measures and ensuring what we are measuring has meaning for the patient as well as for clinical relevance. In 2016/7, I developed a respiratory outcome measure (The Great Ormond Street Respiratory Score for SMA1) which measures not only ventilation but also dependence on respiratory physiotherapy techniques. This was first presented as a poster at the World Muscle Society Conference in 2018. At this conference, I was awarded the Lea Rose Award for best SMA presentation. This work is ongoing and since implementation it has enable better consistency or care with more measurable respiratory function. The longitudinal data was recently presented as a poster at The World Muscle Society Conference in Charleston in 2023 and published as an abstract in Neuromuscular Disorders.

I have continued to develop the research aspect of respiratory management of neuromuscular patients and have recently started a focused research position in this area. This new role has been funded to develop and improve national and international respiratory neuromuscular networks as well as to continue to develop respiratory outcome measures to better monitor these highly complex children. A comprehensive research doctoral training is an essential next step for my future clinical academic pathway as an expert in this field.

I have developed international networks within the neuromuscular respiratory community, including North America, Canada, Europe and the Middle East. I have worked on international standards of care projects (Lillien et al, 2022) as well as being involved in creating a global respiratory measure to help us manage these patients with a greater agreement regarding best patient care (Mayer et al, 2023). This has enabled improved consistency of care as well as improved adherence to standards of care.

My further commitment to research is demonstrated through my successful application for a competitive BRC-NIHR funded internship at Great Ormond Street Hospital, which I am using to develop a robust research protocol and fellowship application to undertake doctoral study. The DACF is an essential step to consolidate, formalise and expand on my research skills and knowledge providing a foundation to embark on a clinical academic career. It would provide me with the tools to develop my understanding of research design and methodology as well as enhancing my statistical analysis skills. It will also enable me to have focused time on PPIE to ensure patient focused outcomes alongside improving physiotherapy care and treatment to benefit children with neuromuscular conditions.

Teaching and training

I view my teaching and training of others as a key component of my current role and as I develop to become a leader in my field. Throughout my career I have provided formal and informal training to other professionals as well as students. I am committed to developing the next generation of physiotherapists and want to ensure high quality knowledge, translation and supervision to these aspiring allied healthcare professionals. They should be able to question practice and progress as a direct result. It is vital to teach with the importance of evidence-based practice and this ensures I remain up to date with current research. I have also educated peers on the importance of data collection, partaking in audits and service evaluations as well as quality improvement projects. My role and skill set have enabled me to teach at a high level on both the national and international stage and I hope to continue to educate as well as develop clinicians as well as professionally.

I am a highly motivated clinical specialist respiratory neuromuscular physiotherapist and my career has enabled me to become a recognised expert in my field, develop highly specialised clinical skills, lead on service developments, teach internationally, participate in and create my own research ideas as well as remain driven to achieve. With a demonstrated commitment to research activity and a drive to improve, I feel I am well placed to undertake an NIHR funded doctoral fellowship.

Plain English Summary (word count 600 – current 600)

Background:

Spinal Muscular Atrophy (SMA) is a life-limiting condition, mostly affecting children, which weakens muscles, including those needed to move limbs, swallow and cough. Mucus builds up in patients' lungs, becoming thick and sticky, causing difficulty breathing. This can lead to chest infections, causing hospital stays and admissions to intensive care. Quality of life and life expectancy is affected. Physiotherapists work with patients and parents to prevent them from becoming unwell. One way is by using a cough-assist machine.

The physiotherapist sets the cough-assist machine on the right settings. The patient temporarily wears a mask over their nose and mouth, with air being blown into their lungs to help them take a big breath in, followed by creating a vacuum to move the secretions out to mimic coughing.

With SMA, we don't understand enough about what happens to the child's airway whilst breathing, during coughing and when using the cough-assist machine. This makes it hard to choose the correct settings on the machine and professionals don't always agree on them.

This study will help us understand more about this and improve children's health, preventing distressing hospital admissions for children and families.

Research aims and design

There are six stages to this study.

1. *Review literature, including other research studies, about respiratory management for children with SMA and their quality of life.*
   This will help ensure our study takes account of any new knowledge or developments.

2. *Understand patients', parents', and carers' views on day-to-day respiratory management.*
   Using SMA UK charity networks to tell families about the project, they will be invited to take part in focus groups (meetings where experiences are shared). This information enables better understanding of the issues which are important to children and families and help create an assessment shaped by patients.

   For the aims 3-6, we will invite children with a diagnosis of SMA under the care of Great Ormond Street Hospital to take part.

3. *Understand what happens to airways of children with SMA when they breathe and use cough-assist machines.*

Using a small (2mm) camera in a long thin tube, placed through the nose, we will look at the airway (This will not hurt). Two recordings will be made: one with and one without the cough-assist machine.

4. *Improve effectiveness of children's coughing by making changes to cough-assist machine settings*
   This involves the equipment as described above. We will adjust the cough-assist machine settings (small changes to air being provided and to cough strength) to:
   - Re-assess the airway and cough effectiveness

5. *Assess breathing patterns of children when using cough-assist machines on initial and adjusted settings.*
   Patients will lie on a bed, breathing normally, with markers on their chest, whilst watching television. 8 cameras pick up signals from the markers and convert the information into the child's breathing pattern. Cough-assist machines will be used on original and adjusted settings evaluating a difference in breathing patterns.

6. *Understand patients', parents' and carers' experiences of taking part in the study and its impact on respiratory care going forward.*
   We will ask children and families who take part in the assessments of airways and breathing to answer a short questionnaire (this will be adapted for children so that they can share their views). This will help us learn more about these tests and how useful they are to patients, to help clinical staff decide on the best way to manage care and treatment.

Dissemination

The study will be supported by a parent steering group. It will be shared through presentations at meetings and conferences (UK and internationally), parent support groups/charities and submissions to peer-reviewed medical/physiotherapy journals.

Scientific Abstract (word count 500)

Detailed Research Plan (word count 5000 – current 4872)

What is the problem being addressed?

Spinal Muscular Atrophy (SMA) is an autosomal recessive neuromuscular disorder caused by the deletion of the survival of motor neuron protein 1 (*SMN1*), responsible for 90% of SMA production in healthy individuals (Finkel et al 2017). In humans, a second gene, survival motor neuron 2 (*SMN2*) exists which alone produces insufficient protein to maintain muscle function. Due to the lack of SMN protein, SMA is characterised by the degeneration of the anterior horn cells of the spinal cord resulting in muscle atrophy and proximal muscle weakness (Edel et al 2022).

It is pan-ethnic with an incidence rate of 1 in 10,000 live births. In Europe the carrier frequency of 1:41. SMA can be divided into different subtypes, depending on severity and age of onset. Copy numbers of the *SMN2* gene is thought to correlate with subtypes, with the higher the copy number, the milder the phenotype as it is associated with more SMN protein being produced (Arnold et al, 2015, Pechmann et al 2018, Talbot and Tizzano 2017).

There are 5 different types of SMA, ranging from the most severe SMA0 to the adult form SMA4. Infants with SMA0 often do not survive the first few months of life and are born with limb restrictions, cardiac and respiratory complications. (Bowerman et al 2017)

SMA1 is the most common and severe form of the disease. Children with SMA1 characteristically present prior to six months of age with progressive muscle weakness, bulbar dysfunction, chest infections and respiratory failure. Progressive respiratory failure is a major cause of morbidity and mortality. SMA1 can be further sub-grouped into types A, B or C. This is a grouping of severity, with type 1A being the most severe. (Baranello et al 2021). Historically, children with SMA1 would have died before the age of 2 years. Over the past 15 – 20 years, the standards of care, supportive therapies and more recently disease modifying therapies have driven a more interventional approach and have improved life expectancy. (Kolb and Kissel, 2015, De Sanctis et al 2018) SMA1 has shifted from a fatal diagnosis to a chronic life-threatening condition (Ngawa et al, 2023). Many children are now living with the disease, attending school and participating in the wider community. SMA 2 and 3 tend to have less respiratory complications, however the spectrum of the disease means any child with SMA may have respiratory compromise.

Children with SMA1, 2 or 3 can struggle with secretion clearance due to a weak and ineffective cough, resulting in mucus plugging, impaired mucociliary clearance and eventually lobar and/or lung collapse (Hov et al, 2018). Secretion management and respiratory physiotherapy are key to preventing respiratory infection, reducing hospital admissions and improving life expectancy (Finder, 2010). These patients require regular review with a specialist physiotherapist and individualised treatment programmes, which include one innovative treatment; the mechanical insufflation-exsufflation device (MIE), commonly called a cough assist machine (Stehling et al 2015). Often, due to their over-riding muscle weakness, an MIE device is a vital treatment to aid survival, as recommended as part of the internationally agreed standards of care (Finkel et al 2018).

With disease modifying treatments available for SMA, which focus on improving motor and respiratory function, ensuring that respiratory management is optimal is a vital part of the patient's care (Talbot and Tizzano 2017) MIE is utilised by clinicians across the United Kingdom and internationally for these patients (Hov et al 2018). MIE provides a breath in which should be large in volume using positive pressure, followed by the cough stage which utilises negative pressure to move secretions higher in the oropharynx so they can be cleared effectively. Setting the optimal pressures on a MIE device has long been a topic of discussion, clinicians are required to use observation and patient feedback to establish appropriate settings and there is no clear guidance as to how settings should be chosen (Hov et al 2023).

Patients who have neuromuscular disease with bulbar dysfunction can be challenging to treat with MIE. Bulbar dysfunction results from neuron degeneration in corticobulbar and brainstem pathways, which is a key presentation, specifically in SMA1A and SMA1B (Andersen et al 2021). Research on upper airway management and response to mechanically assisted cough in another similar motor neurone disease, amyotrophic lateral sclerosis (ALS), that affects an adult population, has shown that in individuals with poor laryngeal control, therapeutic pressures with MIE may cause a disadvantageous laryngeal response, preventing the lungs from filling with air and consequently reducing cough efficacy. Trans-nasal fibreoptic

laryngoscopy (TFL) was used in this patient population to view the upper airway and assess the effect of the MIE, which was well tolerated and able to monitor responses throughout all phases of the cough (Andersen et al 2017). To date, we do not know how the upper airway of children with SMA respond to MIE. Further research is needed to be able to provide optimum evidence-based pressures on MIE and understand the upper airway response to these pressures.

Why is this research important in terms of improving the health and/or wellbeing of the public and/or patients and health and care services?

It is well known that a diagnosis of SMA is linked to respiratory complications which are the major cause of hospitalisation, morbidity and mortality (Edel et al 2022). Successful implementation of respiratory standards of care has improved life expectancy. A key part of this is adequate airway clearance which is a vital part in improving survival, especially the correct application of MIE (Finder 2010). Information regarding specific settings and the effect of these on the airway is largely unknown and is likely to have a large impact on the effectiveness of the treatment. This research will explore these factors, improving the understanding of what occurs during mechanical insufflation exsufflation to inform clinicians of best practice for these patients, helping improve their airway clearance as well as their health and quality of life.

Improving respiratory outcomes

Detailed study of the upper airway during mechanical insufflation exsufflation will allow us to ensure effective delivery of pressures, gaining maximum benefit to the patient. In turn, this would improve lung expansion and reduce potential areas of lung collapse, decreasing the risk of chest infections and less hospital admissions which have a direct effect on morbidity and mortality in this patient cohort.

Reducing hospital admissions

Despite disease modifying therapies, there remains a high rate of hospitalisations due to respiratory complications in SMA patients. In a small cohort study in the north of England, it was found that approximately 20% of a child's life was spent in hospital, mainly in an intensive care or high dependency unit. The estimated cost of this care was £2.2M (Ali et al 2019). Similarly, in the United States, each admission for a child with SMA1 costs appropriately $50,190 and the cost may be underestimated given they require an average of 4.2 hospitalisations annually. The hospitalisations are also reported at 4.4-fold longer with multiple concerns on admission (Cardenad et al 2019). In addition to this high-cost impact on the National Health Service, there is a large physiological, psychological, and economic burden to the patient and family. It also impacts on the child's education and their social development.

Improving patient/carer quality of life and well being

There is limited information regarding quality of life in terms of respiratory burden of care for patients and families. More recently studies have begun to look at user perspective for mechanical insufflation exsufflation in relation to effectiveness. However, there is no validated quality of life measure which focuses on respiratory burden in neuromuscular conditions. This research will explore the key factors families and children identify as significant for their experiences to help identify key components regarding quality of life and wellbeing for respiratory management.

Supporting development of care pathways

The standards of care (2018) highlight mechanical insufflation exsufflation to be the gold standard for airway clearance for SMA (Finder 2010, Hov 2018, Edel 2022). However, limited data are collected regarding settings and use. Due to funding restraints, there is also variable access to these devices across the United Kingdom. This research will develop setting parameters for children with SMA to help guide settings with the view to create a national pathway or clinical guideline to help future clinicians and patients.

Review of existing evidence – How does the existing literature support this proposal?

With the introduction of 3 disease modifying therapies (DMT) (Baranello et al 2021) and more interest into the respiratory outcomes, this proposal is key. The current standards of care (Finkel et al 2018) were written prior to DMTs becoming available. The life expectancy for these patients is increasing and it is vital their respiratory management is optimum.

There is existing literature reviewing the upper airway in adults ALS (Andersen et al 2021) and a single case review looking at non-invasive ventilation in one child with SMA1 (Vollsaeter et al 2021). We are unable to draw specific conclusions regarding a larger cohort of SMA1 patients and the airway clearance for the patients is highly complex with a high impact for patients and clinicians.

Use of TFL

It has been highlighted mechanical insufflation exsufflation is not effective for all patients (Andersen et al 2021). Poor bulbar control may cause obstruction to air flow through the larynx. The larynx consists of 3 main layers along with mucosa, ligaments and muscles that are responsible for the abduction (opening) and adduction (closing) of the airway and it is important that these work in a co-ordinated way for effective use of mechanical insufflation exsufflation. Current literature in ALS suggests that an individual approach to setting mechanical insufflation exsufflation is important and highlight treatment failure is likely due to laryngeal adduction during insufflation (Andersen et al 2017, 2021). This response precludes air filling of the lungs during insufflation, causing discomfort and subsequent inefficient exsufflation. The clinical translation in to SMA remains a question. Utilising TFL is the gold standard test for visualising the upper airway (Halvorsen et al 2017). It is well tolerated and has been used with success (Andersen et al 2013). It enables us to separate out the larynx into supraglottic, glottic and hypopharyngeal and document to sequence of the airway abduction and adduction.

The use of TFL was discussed during patient and public involvement activities. The parents initially expressed concern over the use of TFL, however when the size was highlighted as being smaller than a suction catheter they use for day to day use, there was consensus that this would be acceptable and well tolerated by children. They have suggested including patient videos and step by step guides to help the children understand the test. This is something we will look to develop as part of the proposal.

Opto-electronic plethysmography (OEP)

Respiratory lung function assessments in SMA 1 children have always been challenging as it usually involves invasive or volitional tests. Patients with SMA1 often have more rapid and shallow breathing with paradoxical thoracoabdominal movements and a bell shaped chest (LoMAuro et al 2018). OEP has been utilised by other researchers to investigate the total and compartmental chest wall volumes. It is able to measure quiet breathing as well as inspiratory capacity and voluntary cough. One advantage of utilising the OEP is that it can be performed in young babies as it does not require cooperation. The use of OEP in this young population is particularly suitable as there is a lack of reliable outcome measures at this age. It has been shown that SMA children have a degree of asynchrony with their breathing as well as a more restrictive chest wall. OEP has been shown to be safe and well tolerated by patients of all ages (LoMauro et al 2014). Further research looking at other patients with neuromuscular disease have shown that a single treatment with MIE can provide beneficial changes to the breathing pattern by improving short term dyspnea, although there were no changes in lung volume recruitment or cough peak flow. (Cesareo et al 2018)

Patient experience and quality of life

There has been limited work looking at this in the respiratory neuromuscular paediatric population. There are many questionnaires focusing on quality of life, however, these are more specific to other conditions or general disabilities. The current work highlighted on the COMET initiative is focused on adults in the neuromuscular community. Alderhey Children's hospital have validated a quality-of-life questionnaire for

children with neuro-disability (Trinick et al, 2015), which is a useful tool for the benchmark of the questionnaire this project is looking to develop.

There is a clear need to improve the respiratory management of children with SMA and to explore the interaction of mechanical insufflation exsufflation with the upper airway and its impact on lung volume and cough peak flow.

What is the research question / aims and objectives?

- Research Question:

Can we utilise trans-nasal flexible laryngoscopy to optimise the upper airway response to mechanical insufflation exsufflation in children with spinal muscular atrophy and what are the subsequent effects on respiratory function and to patient respiratory care and experience?

- Aims/Objectives:

Aim: To understand the reason why there is variability in effectiveness and tolerance of children to mechanical insufflation exsufflation

Objectives:

1. To complete a comprehensive systematic review of the literature using PRISMA-P protocol
2. To assess if TFL is feasible to use for airway assess and during MIE with children with SMA
3. To describe and evaluate the upper airway response patterns with mechanical insufflation exsufflation
4. To evaluate whether changes in mechanical insufflation exsufflation settings can alter the response patterns seen in the upper airway
5. To assess changes in spontaneous respiratory patterns with optimised mechanical insufflation exsufflation settings
6. To understand patients, parents, and carers views on day-to-day respiratory management including the use of their cough assist machine.
7. To understand patients, parents and carers tolerance and acceptance of new assessment techniques and their views on the contribution to respiratory care.

Project Plan

A prospective interventional cohort mixed methods study has been designed to investigate upper airway response to mechanical insufflation exsufflation in children with spinal muscular atrophy. This study will include 5 work packages. The initial work package will be a systematic review of the literature. This will inform the following 3 work packages collecting quantitative data focusing on the specific upper airway anatomy and response to mechanical insufflation exsufflation. Work packages 2, 3 and 4 will be run concurrently. Work package 5 will focus on qualitative data aiming to gain specific feedback from patients

and parents regarding the acceptability of the techniques used as well as their experiences of respiratory management as it impacts on their day to day lives.

Work Package 1:

Systematic literature review

Using the PRISMA-P protocol, the literature related to SMA respiratory management, trans-nasal flexible laryngoscopy, opto-electronic plethysmography, and respiratory quality of life outcome measures will be systematically reviewed. Study types will include but not be restricted to, randomised control trials, cohort studies and case studies. I will search a range of databases including PUB MED, Research gate and web of science with full text articles being obtained. I will develop a search strategy with an information specialist, but terms would include; (Spinal muscular atrophy AND/OR respiratory management AND/OR respiratory physiotherapy AND/OR airway clearance techniques AND/OR mechanical insufflation exsufflation) AND/OR (tran-nasal flexible laryngoscopy AND/OR neuromuscular disease) AND (opto-electronic plethysmography AND/OR neuromuscular disease) AND/OR (respiratory quality of life AND/OR neuromuscular disease)) will be used.

Work Package 2:

This work package will be inductive research, gathering information regarding patient and parent experience related to respiratory management which will be incorporated into a questionnaire design regarding quality of life for children and families with SMA.

- Families and children from the national Spinal Muscular Atrophy network as well as those captured via advertising through the charity's networks, will be asked to take part in focus groups regarding their respiratory management and quality of life/burden of care. The reason for this population for data collection is to enable feedback to be gained from those patients outside of Great Ormond Street Hospital. There is a variation in the standards of care throughout the United Kingdom and it is important to capture how that impacts respiratory health. Focus groups will be utilised to enable discussion between patients and families.
- A topic guide will be developed combining preliminary and ongoing PPIE work and the literature as well as current respiratory outcome measures (Liverpool Respiratory Symptom Questionnaire as well PedsQL Neuromuscular Questionnaire). The topic and content will be outlined, taking in to account the participants point of view. Questions will be open and have prompts when needed.
- Questions will include:
    - What are family experiences of their respiratory management?
    - What is the impact of Mechanical Insufflation Exsufflation
    - How do families and children feel about their respiratory management plan?
- The focus groups will be completed prior to other work packages. There will be a range of data collection tools to ensure the view of the family and the child are captured. This is likely to be in the form of picture cards/talking mats and emoji scales.

Inclusion Criteria

All patients and families within the UK SMA reach network would be eligible for completing the focus groups regarding their experience and understanding of their respiratory management.

Data Collection

Thematic analysis will be used to analyse the data collected from the semi structured interviews and focus groups.

Work Package 3:

Study design:

- This will be an observational cohort mixed methods study focused on baseline characteristics of the airway. Written consent will be sought from parents and/or patients.
- Utilising existing databases, baseline characteristics will be obtained for all patients:
    - Age/Date of birth,
    - Sex,
    - Growth centile (Weight/Height),
    - Copies SMN,
    - SMA Treatment and age at treatment,
    - Non-invasive ventilation (NIV) user/hours of ventilation,
    - Other chest physiotherapy techniques/nebulisers,
    - Other medications,
    - Compliance data to MIE/compliance to NIV,
    - Existing secretion management plan/baseline MIE pressures and start date,
    - Functional ability/motor function score (CHOP (Childrens Hospital of Philadelphia Infant Test of Strength in SMA)/RHS (Revised Hammersmith Scale))
    - Scoliosis (Cobb angle) +/- surgery
- Video-recorded trans-nasal flexible laryngoscopy (TFL) will be used to visualise laryngeal anatomy at baseline and during baseline individualised patient mechanical insufflation exsufflation settings (as determined prior to start of the study)
- The baseline upper airway will be observed via TFL and the different levels of the upper airway (Glottic, Supraglottic and Hypopharyngeal) would be scored (0 open, 1 collapse).
- Following this, the baseline individualised patient mechanical insufflation exsufflation settings will be assessed.
- The laryngoscope will pass through a modified interface which will attach to the mechanical insufflation exsufflation device, this will then be placed on the patients face.
- The laryngoscope, which is 2.8mm in diameter, will then be supported and introduced to the patient trans-nasally.
- The examination will be visualised on a screen with continuously running video recordings with the endoscopic video from the TFL and an external video camera to the control of the MIE device, allowing for documentation of the phases of insufflation and exsufflation.
- The TFL will be connected to the video which will monitor the upper airway. The different levels of the upper airway (Glottic, Supraglottic and Hypopharyngeal) would be scored (0 no change, 1 collapse). The result would be the outcome of what happens to the airway under direct visualisation.

- The other parts in the equipment set up would be; a camera set up to monitor the pressure, another to monitor the patient, and a pneumotach and pressure transducer to give an exact pressure reading from the MIE device.
- Inter-rater reliability will be measured through review of recordings by a blinded assessor.

Power Calculation

- From our pilot data, it has been identified that a clinically significant meaningful change in respiratory insufflation volume is 1.5mls/kg when adjusting mechanical insufflation exsufflation pressures. With a standard deviation of 3.2, we would need 36 patients to account for this difference as statistically significant with 80% power and 5% significance. To allow for 10% attrition (patient drop out, illness and death), we would look to recruit 50 patients from all functional groups to ensure we recruit at least 36 patients. This power calculation applies to work packages 3, 4 and 5.

Sampling frame

The sampling frame will focus on current patients. It will be based on the following inclusion criteria:

- Inclusion:
    - Genetic confirmation of spinal muscular atrophy
    - Use of mechanical insufflation exsufflation for airway clearance for at least 3 months
    - Care under the respiratory physiotherapy team at Great Ormond Street Hospital
    - Age 0 to 18 years
- Exclusion:
    - Patients who are on a palliative care pathway
    - Patients who have transitioned to adult services

Primary Outcome: Airway status at rest and airway status at baseline mechanical insufflation exsufflation settings. An open airway will be scored as 0 and a closed airway will be scored as 1.

Data collection and analysis

Data collection with be completed over 3 months. This is with the goal of testing 16 patients per month, with a maximum of 2 patients per day.

Data will be collected on an existing evaluation form based on the work by Andersen et al (2013) to allow for review of each level of the larynx and to analysis if it is open or closed. The evaluation form will also allow for comments as well as documentation to indicated at what stage of the mechanical insufflation exsufflation cycle the view was made at. The videos will be screened and analysed.

The anonymised videos will be upload to a secure online platform to allow for inter-rater reliability testing.

Data will be recorded in REDcap, coded and transferred to R for statistical analysis. The baseline characteristics will be analysed using means and standard deviations, or number and proportion as

appropriate. Between group demographic analysis based on functional levels will be analysed using ANOVA, Kruskal Wallis tests or chi square test.

The airway status will be analysed as open or closed at each level of the upper airway and we will compare the proportion of open airways at each level by functional groups (non sitters, sitters and walkers combined) and then analysed utilising the Chi squared or Fishers exact test.

In addition, we will compare the lowest open level between functional groups using Mann Whitney tests.

Work package 4:

This work package will run concurrently with work package 3.

This is an interventional pre and post study with no control group, examining the effect of changing mechanical insufflation exsufflation settings on the upper airway. Patients will start on the pressures identified in work package 3 and following a protocol driven stepwise approach the settings will be altered to obtain the highest pressures with an open upper airway. This will provide rich data as to characterise the upper airway at different levels of positive and negative pressure. This data collection will occur at the same time point at work package 3.

Additionally, a short series of questions will be developed and will ask the patient and family about respiratory health over the last 6 months.

- Patient will start on baseline mechanical insufflation exsufflation settings. The protocol will be followed to obtain best pressures on MIE with an open upper airway.
- The laryngoscope will pass through a modified interface which will attach to the mechanical insufflation exsufflation device, this will then be placed on the patients face.
- The laryngoscope, which is 2.8mm in diameter, will then be supported and introduced to the patient trans-nasally.
- The examination will be visualised on a screen with continuously running video recordings with the endoscopic video from the TFL and an external video camera to the control of the MIE device, allowing for documentation of the phases of insufflation and exsufflation.
- The TFL will be connected to the video which will monitor the upper airway. The different levels of the upper airway (Glottic, Supraglottic and Hypopharyngeal) would be scored (0 no change, 1 collapse). The result would be the outcome of what happens to the airway under direct visualisation.
- The TFL will be left in situ as the mechanical insufflation exsufflation pressures are amended with the view to obtain the optimal settings for that patient.
- These settings will be adjusted depending on the view seen at baseline. If the airways are open at baseline, the MIE device settings will be increased in 2cmH2O increments until airway closure is noted. The settings will then be reduced to the last open level documented. If the airways are closed at baseline, the MIE settings will be decreased in 2cmH2O increments until airway opening is seen. These settings will then be documented as the patients optimal settings.
- The other parts in the equipment set up would be; a camera set up to monitor the pressure, another to monitor the patient, and a pneumotach and pressure transducer to give an exact pressure reading from the MIE device, including peak cough flows.
- Inter-rater reliability measured through review of recordings by blinded assessors.
- All patients who are able to communicate will also be asked to report the level of comfort of the settings using a visual analogue scale.

Sampling frame:

This will be a follow-on plan from work package 3 and sample size will remain at 50 patients. All patients from work package 3 will be eligible to take part. Written consent will be obtained from parents or patients if appropriate.

Exclusion:

- Non tolerance of TFL from work package 3.

Primary Outcome measure: Maximal airway pressure from MIE with all levels of airway open compared between functional groups.

Secondary Outcome measure: Qualitative data on respiratory health and comfort and tolerance to settings

Data Collection and analysis:

Data collection will occur at a range of settings on the mechanical insufflation exsufflation device. This will occur during one session unless the patient requires a rest period.

This data collection will occur at the same time point as work packages 3 and 4.

The airway status will be analysed as open or closed at each level of the upper airway at the different settings with clear documentation of the maximal settings. The maximal setting between groups will then be analysed and compared utilising a T-test or Mann-Whitney test.

The cough peak flow from baseline settings and maximal modified settings will be compared and analysed with paired t-test or Wilcoxon matched pairs test.

The data will be analysed for relationships between age and SMA treatment plans.

Work package 5:

This will focus on reviewing the lung function of patients following adaptation of the mechanical insufflation exsufflation settings.

Utilising OEP with the patient in the supine position, markers will be placed in specific locations throughout the thorax and abdomen.

Once the patient is settled and comfortable, data will be obtained whilst the patient is completing spontaneous quiet breathing.

They will be asked to complete a natural cough, followed by a treatment session with their baseline settings of mechanical insufflation exsufflation and a second period of quiet breathing.

Lastly, they will be asked to complete the treatment session with the newly adapted settings.

They will finish with a period of quiet breathing.

Sampling frame:

This will be a follow on plan from work packages 3 and 4, completed at the same visit after a short break. All patients from work packages two and three will be eligible to take part and the sample size remains at 50 patients. Written consent will be obtained from parents or patients if appropriate.

Exclusion:

- Non tolerance of TFL from work package 3 or 4
- Unable to complete the period of quiet breathing

Data Collection and Analysis

The data will be obtained through the computer programme EVO and saved under the code from work plan 2 and 3. It will be analysed using another programme called SMART TRACKER and the data produced will include: tidal volume (TV), respiratory rate (RR), Forced Vital Capacity (FVC), forced Expiratory Volume (FEV1), Total lung capacity (TLC), compartmental chest wall volumes and Cough peak flow (CPF).

Tidal volume data from the original mechanical insufflation exsufflations settings will be compared to the adapted settings and will be analysed using a paired t test or a Wilcoxon matched pairs test. From initial pilot data, it has been identified a meaningful change in respiratory insufflation volume is 1.5mls/kg when adjusting mechanical insufflation exsufflation pressures.

Work Package 6:

This work package will be divided in to two main parts. The first part will comprise of one-to-one meetings with parents and patients regarding the acceptability of introducing testing with TFL and OEP. This will feed into the questionnaire design which was started in work package 2. The second part will be a pilot of the questionnaire which will be conducted anonymously. The results from the questionnaire will be analysed to give information on patient experience and quality of life.

- This will focus on families involved in Work Packages 3-5 and they will be asked in one to one meetings regarding the new assessments utilised in the project. Questions will include:
    - What do they think about the new assessments?
    - How did the new settings on their mechanical insufflation exsufflation compare to their old settings?
    - How did the Opto-electronic plethysmography markers and cameras feel?
    - Does your secretion clearance feel better/worse/no different with the new settings compared to the old settings?
    - Would this fit in to their clinical assessments?

Once the one-to-one meetings have been completed, a respiratory quality of life questionnaire will be developed to be used alongside current clinical assessments. Professor McNamara at Alderhey will be a collaborator on this work package to further develop the questionnaire.

- The second part will be a pilot of the questionnaire with all families involved in work packages 3-5 at a six month follow up appointment.

Further work required to validate the questionnaire will be focused on in a Post Doctoral project.

Inclusion Criteria

All patients and parents that have been identified in work package 3-5 would be eligible for one to one interviews post intervention to understand their experience and understanding of the novel respiratory assessments as well as the 6 month follow up appointment.

Written consent will be obtained from the parents and patients. For children, age specific consent or assent will be developed.

Data Collection

Thematic analysis will be used to analyse the data collected from the interviews.

The questionnaire will be used to collect respiratory quality of life data linked to this proposal and a scoring system will be devised.

Control Group

There will be no control group for this proposal. There would be ethical issues with applying mechanical insufflation exsufflation to children that do not have a requirement for this device and are not acclimatised to the device. This is likely to cause them to have increased resistance to the machine and give unreliable results. This is a prospective observational study of a new intervention.

Output, anticipated impacts and dissemination

Output

This proposal will help to improve respiratory management in patients with SMA and ultimately their quality of life, by identifying appropriate airway clearance parameters to use in their secretion clearance management.

Improved clinical decisions

A constant challenge in respiratory management is how clinicians should be setting up mechanical insufflation exsufflation devices and the pressures needed for optimal airway clearance. Utilising TFL will give us a range of parameters for MIE settings in this cohort and help guide care.

For the first time in children with SMA, we will have direct measures of lung volumes utilising the OEP as well as access to information regarding the upper airway of these patients.

Creating a respiratory quality of life questionnaire for SMA will be the first step to a validated tool to better understand the impact of respiratory management on these patients', families' and carers'.

Future study designs

Work would be undertaken to validate the SMA quality of life questionnaire we will create during this project.

In future studies, we would complete a comparison study between patients who follow standards of care for SMA and patients who have had their MIE adjusted under direct visualisation with TFL. We would utilise the quality of life questionnaire and OEP to understand if there is a difference between groups.

Further development of utilisation of the techniques in other neuromuscular and chronic long-term conditions is needed. It is unknown how the upper airway is affected during use of mechanical insufflation exsufflation in most conditions as well as the impact on lung function parameters. It would also be important to allow patients outside of Great Ormond Street to have access to further studies.

Anticipated Impact

By completing this project, personalised settings for mechanical insufflation exsufflation would help reduce the risk of dynamic airway collapse during treatments. The potential impact of this research are far reaching. It would allow future prescribing of MIE to be more specific, which would improve secretion clearance, reduce the risk of infections and potentially prevent admissions. This would also reduce the potential disruption to the patient, parents and other family members. There would be potential to develop and led a complex airway clinic to improve patient care and generate income for the trust.

It may also translate to other conditions who require MIE, helping more patients. This project will also provide evidence on the safety and efficacy of mechanical insufflation exsufflation in the paediatric population. It will enable the professional community to come to a consensus on MIE setting prescriptions where there is currently a gap in understanding.

Dissemination

Conference presentations and publications

Conference presentations and publications are planned for appropriate time points throughout the fellowship, as well as reaching out to charity partners within SMA to engage families from a diverse range of backgrounds are able engage in patient facing presentations. Presentations and/or handouts can be translated in to a variety of languages if required.

Conferences will be targeted to reach families and appropriate multi-disciplinary audiences and include:

- International Conference on Home Mechanical Ventilation and European Respiratory Care Association Congress JIVDERCA,
- World Muscle Society,
- SMA Europe,
- CureSMA,
- European Respiratory Society.

A minimum of four journal articles are planned with at least one being open access. Targeted journals will include; Thorax, American Journal of Respiratory and critical care medicine and Neuromuscular disorders. Intended titles would be:

Is trans-nasal flexible laryngoscope a feasible tool to use in SMA?

Can children tolerate high pressures on mechanical insufflation exsufflation?

What factors influence respiratory quality of like in children with spinal muscular atrophy?

Does mechanical insufflation exsufflation have an impact on the upper airway in children with spinal muscular atrophy?

Parents/Patients/Charity/Support group dissemination

During project planning parents and patients have highlighted the importance of feedback. They have provided feedback for the study design and will continue to be involved in the steering group. This will ensure the findings are widely accessible and all target audiences reached, maximising the impact.

SMA UK and MD UK will support dissemination by sharing findings on their website and social media.

Information and results will be appropriately fed back to parents and patients both during and at the end of the study.

International influence

A number of international charities and networks exist for SMA and it will be key to link with these to ensure international output of the study results.

Feedback to parents/carers/patients and the general public

During planning, parents and patients have highlighted the importance of communication and feedback. They provided useful insight into the preferred dissemination methods and this plan has been developed with their recommendations.

- Lay summary for parents/carers and patients in a leaflet and infographic format via coffee mornings, notice boards and parent conferences
- Online reports for relevant charities and GOSH websites

- Participation at GOSH/BRC research events

Project management

A detailed management plan will be developed in collaboration with all supervisors, incorporating the study work packages, academic training, and clinical development. It will be reviewed regularly to ensure the project is on schedule and no new issues have developed.

The project will be overseen by my primary doctoral supervisor, Professor Giovanni Baranello. Further academic supervision will be provided by Dr Colin Butler and Dr Tiina Andersen.

Practice Supervision will be provided by Dr Martin Samuels and Paula Kelly

A meetings schedule is included in the Gantt chart.

The applicant will meet monthly with the primary academic supervisor Professor G Baranello and also with Dr C Butler and Dr T Andersen. Joint research supervision with all supervisors will occur every 6 months and at key points within the project. Additional contact will be sought as necessary in between scheduled meetings.

Formal clinical supervision will occur monthly with either Dr C Butler or Dr M Samuels, and quarterly online meetings will be scheduled with Dr P Kelly and Dr T Andersen, in addition to face-to-face supervision during the planned visits.

Steering group meetings will be included at key points during the project to advise on study management and also ensure parent/carer involvement. Support and guidance for management of the data set and statistics will be provided by statistician Ms D. Ridout.

Financial management will be undertaken by the finance department at GOSH. The applicant will meet monthly with the finance team and keep an accurate personal record of expenditure.

Collaborations

Collaborations within the NHS: Paul McNamara at ALderhey hospital

Collaborations from Industry: ChilliPharm, Breas, BTS, Storz

Ethical Considerations

Full ethical approval will be sort for work packages 2, 3, 4, 5 and 6 in months 1-7. The supervisory team will be fully involved with ensuring ethical approval is secured in a timely manner.

TFL is an invasive procedure but with minimal risk factors. Consent will be sought from parents or patients if they are Glick competent. We will utilise pictural diagrams and videos to ensure they understand the procedure.

OEP is non-invasive but does require the parent to stay with the children to keep them relaxed. Pictural diagrams will again be used to explain the procedure and ensure the parent and child understands.

Data protection laws will always be adhered to. Confidentiality during each work package will be assured, including participants are not identifiable during dissemination.

Maintaining Anonymity and Confidentiality

Data will be anonymised
An encrypted digital recorder will be used for the interviews
Data will be presented sensitively to avoid participant identification

Data Protection and Security
Data will be handled in accordance with the Data Protection Act 1998 and the specific NHS trust policy
Data and the electronic workspace will be stored securely using a password protected computer and encrypted data stick/hard drive
Only the research team will have access

Informed Consent
Clear information provided on the purpose of study, research team, funding, how data will be used, and participation requirements
Voluntary participation
Written consent for all workplans

Researcher/participant Relationship
Due to my dual role as researcher and physiotherapist, and the potential for prior relationships with the participants a power dynamic may exist which could impact on the success of the interview. Steps to ameliorate this include:
Engagement in self-reflexivity
Presenting self objectively and neutrally

Success Criteria

Patient and Public Involvement and Engagement

This research proposal was developed in collaboration with parents, carers and patients of Great Ormond Street as well as charity partners who are linked through the care for children with SMA. This enabled us to gather important information regarding the proposed project, the study design, any concerns as well as asking if they felt the project was worthwhile.

Six parents, three patients and two charity partners were consulted on the plain English summary as well as involved in group and one-to-one discussions using open questions on:

- Is the research question worthwhile and would it help the SMA community?
- What do you think of the study design, including the tests we are using and the structure of the data collection?
- Would you change anything in the overall question or design?
- What concerns would you have regarding the research?
- Could we do anything to make it easier for families to take part?

The Child and Young Person Group for Great Ormond Street Hospital contributed to the preparation of the application but reviewing and advising on the plain English summary of the proposal.

The families involved confirmed the importance of the research are and the questions posed, and their feedback has informed the development of the proposal and design of the study from the outset.

- The families explained it is research that needs to happen as they feel it would be advantageous to helping their children stay out of hospital. One family felt it 'gives us a door to open up and see what is happening inside and that would be the best.' They reported 'we are grateful you are keen to help and look into the respiratory side of SMA'.

Below are some concerns that were raised and the revisions and additions that were made as a result of the feedback:

- Some patients and parents expressed concern over the transnasal flexible laryngoscope. They had concerns over what it would look like, how it would feel and what would happen if the child/parent did not like it. It was suggested that we make a child friendly video/cartoon to explain the procedure step by step. We will add a QR link so the families can watch it in the clinic but also when they go home. Communication will be highlighted, and each step will be explained. At the consent visit, patients requested to have the opportunity to see the equipment that will be used. It will be ensured that it is accessible to a wide range of ages and cognitive abilities.
- Some families were concerned over the level of fatigue the children may feel. They requested that the testing for the transnasal flexible laryngoscope all be completed in one visit with breaks as needed, followed by a longer break and then the opto electronic plethysmography in the afternoon. They expressed a want for testing to be completed in one day with no other booked appointments.
- The parents, families and the charity all stated their interest in receiving feedback about all the elements of the study. They requested appropriate feedback to ensure it was all inclusive.

Ongoing PPIE

Undertaking/Managing the Research

A steering group will be created comprised of 4 parents, 2 ex-patients (patients who have transitioned to adult services) and 2 representatives from charity partners. All parents involved in the development stage have expressed an interest in joining the group. One charity partner has also expressed an interest. Further recruitment will be through email correspondence and public engagement on social media. Anyone interested will be offered a follow up meeting.

Parents/Patients/Charity Partners will have the opportunity to:

- Partake in training for being involved in an advisory group
- Guide the project throughout the research process
- Explore ethical issues
- Assist with plain English summaries/updates
- Provide patient representation, parent perspective and SMA network links.

This group will meet 3 times face to face (once a year) and online groups approximately 5-6 monthly and will be consulted via emails/telephone calls at other key points during the project.

Dissemination

Where appropriate the steering group will be invited to assist with writing final reports/summaries of the results and will be acknowledge for their involvement in the study. Their role in facilitating publication of the findings on charity/SMA specific websites and dissemination through their informal networks will be invaluable and key in ensuring appropriate dissemination to all audiences.

Training and Support

The applicant will attend the INVOLVE conference within the first six months of the fellowship to increase knowledge of PPI and its practical application. When recruiting parents/carers to the steering group a clear description of the roles/responsibilities will be provided, plus the opportunity for discussion. Training needs will be identified and supported by the applicant's current networks (GOSH PPI Lead/ academic supervisors). PPI expenses will be calculated at the recommended INVOLVE rate. As parents/ carers will be brought together in steering group meetings it is acknowledged that they may share the burden of disease, hence psychology support will be available.

Training and Development Programme (word count 678/1000)

Proposed training and development programme
The focus of this training and development plan is to ensure both my academic and clinical needs are met, enabling my progression as a clinical academic leader within neuromuscular respiratory physiotherapy. A comprehensive strategy has been developed including formal academic tuition, specialist training, conferences, and an overseas visit. I have also completed training in preparation for this proposal and project. The training plan is integrated in the Gantt chart.

Academic Training
University College London (UCL) has a world leading reputation and registration on the graduate research development programme will provide access to an extensive range of online resources and taught courses. My training and development plan has been created using the VITAE Researcher Development Framework, which ensures my research needs are met within all four domains. These encompass knowledge and intellectual abilities, personal effectiveness, research organisation and governance, and communication and impact. The formal and additional academic training, outlined below, has been developed to cover all aspects of work package one to five.
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Formal Academic Training as part of the UCL PhD programme

Year 1
- An introduction to R with RStudio
- In a Nutshell: starting an NVivo project
- Introduction to REDCap for research
- Data import and export in REDCap
- Managing Participants in REDCap
- Getting started with your systematic review
- How to do PPI: focus groups and more
- The Writing Series: Academic Writing logic
- Introduction to Research Integrity and Support
- Mixed methods approach in social sciences

Year 2
- Introduction to qualitative analysis: Interviewing
- Introduction to qualitative research method: thematic analysis
- Engagement and Impact: Inclusive practice
- Introduction to writing your PhD thesis.

Year 3
- Academic Career Planning for PhDs
- Networking Skills
- Researchers skills beyond academia: leadership

Additional Academic Training

These additional opportunities will allow me to upskill in specific areas relevant to my project.

Year 1
- Mixed methods systematic review training, University of Leeds, faculty of medicine and health
- Questionnaire Design, online course with national centre for social research
- Statistics courses: Chi-square and Beyond for 2x2 Tables at UCL

Year 2
- Qualitative Research Methods, University of Oxford

I will also have access to additional research support and training through involvement with The Centre for Outcomes and Experience Research in Children's Health, Illness and Disability (ORCHID). This is a dedicated research unit at Great Ormond Street Hospital including clinical academic nurses and allied health professionals (AHP). ORCHID supports nursing/AHP research within the Trust through a drop-in service, writing mentorship and an annual conference.

Clinical Training
Throughout the fellowship my priorities will be to maintain current clinical competence whilst also advancing my clinical knowledge and skills.

Planned training is below:
Year 1
- Development of advanced clinical skills
    - Clinical Training for trans-nasal flexible laryngoscopy with Tiina Andersen
        - Tiina developed the testing procedure and will be the key person in teaching and developing my clinical skills in this area.
        - Two days would be spent observing, followed by six days of hands-on teaching and skill development, and two days of competence sign off
    - Clinical supervision with Colin Butler
    - Clinical time at NMCCC at Queens Square observing their TFL clinic (2 days)
      Purpose of visit
        - Observation of physiotherapy led TFL in practice
        - Collaboration with physiotherapists using TFL
        - Engagement and practical ideas of how to integrate TFL within own clinical practice
- Maintenance of current clinical competence
    - Up to 1-2 days per month carrying complex clinical caseload
    - Providing teaching to the developing neuromuscular respiratory team

Year 2
- Reciprocal visit from Dr Andersen for supervision in own paediatric setting
    - 1-2 days per month carrying complex caseload
- Increased knowledge of clinical specialist/consultant physiotherapy roles; specifically the integration of clinical and academic aspects

Year 3
- Maintenance of advanced clinical skills and competence
    - On-going clinical supervision
- Development of leadership skills
    - Leadership Course: Emerging Clinical Leaders, The Kings Fund, 6 days

Conferences
- International Conference on Home Mechanical Ventilation and European Respiratory Care Association Congress JIVDERCA 2025
    - This International multidisciplinary (MDT) conference will offer clinical learning and networking.
    - The conference only meets every third year
- CureSMA 2025

- This international conference is a family centre conference and would be an excellent opportunity to discuss the study to continue PPIE
- WMS 2026
  - This International conference will be an excellent platform to disseminate study findings to date
- ERS 2027
  - This conference will be an excellent opportunity to disseminate study findings in the final year of the PhD.

Mitigation

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